# Changhao Li

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## RESEARCH INTERESTS

*I work toward **Agentic Recursive Self-Improvement** to advance **AutoResearch** and **Long-Horizon Problem Solving**.*

**Focus:** Harness Evolving; On-policy Learning; Environments & Rubrics; Agentic-training Infrastructure.

## EDUCATION

**Georgia Institute of Technology** Aug 2024 – Present
Ph.D. in Computational Science and Engineering GPA: 4.00/4.00
Advisors: Bo Dai and Chao Zhang

**Tsinghua University** Sep 2020 – Jun 2024
B.Eng. in Computer Science and Technology GPA: 3.89/4.00
Advisors: Jianfei Chen and Jun Zhu

## INDUSTRY EXPERIENCE

**ByteDance Seed** — Student Researcher Aug 2026 – Present
San Jose, CA | *Worked with:* Xin Dong

- **Long-horizon coding agents:** Improving Seed models' behavior through post-training to strengthen performance on long-horizon coding tasks.
- **AutoResearch:** Building agents that identify worthwhile research questions and develop promising ideas, extending the research agenda from task execution to research judgment.

**Qwen** — Research Intern May 2026 – Aug 2026
Seattle, WA | *Worked with:* Guoyin Wang and Jingren Zhou

- **Unified and efficient OPD:** Proposed a unified on-policy distillation framework organizing prior methods around **four design choices**, evaluated **30+ configurations**, and developed a simple, effective, and efficient OPD algorithm.
- **Large-scale OPD:** Contributed to the training infrastructure for **Qwen 3.6–3.8**, implementing OPD algorithms and validating their effectiveness on a **120B-parameter model** for long-horizon coding tasks, including SWE-bench and Terminal-Bench.

**AWS AI Lab** — Applied Scientist Intern May 2025 – Aug 2025
New York, NY | *Worked with:* Elman Mansimov, Elad Ben Avraham, Ron Litman, and Wei Xia

- **Deep Research evaluation:** Co-developed **DREAM**, a reference-free evaluation framework combining tool-based verification and adaptive rubrics to assess factual accuracy, temporal validity, and reasoning quality in Deep Research reports.

## SELECTED RESEARCH PROJECTS

2026 **Agent DAgger: on-policy learning for coding agents** [Paper] [Code]

- **Teacher-interleaved rollouts:** Combined on-policy state coverage with dense expert supervision to reduce covariate shift in long-horizon coding agents.
- **SWE-bench performance:** Achieved **27.3% / 29.8%** task resolution with 4B / 8B students on SWE-bench Verified, exceeding the strongest evaluated post-training baseline by **3.9 / 3.6 percentage points**.

2026 **STACX: infrastructure for end-to-end agentic RL** [Code]

- **Project Lead:** Led overall agentic system planning, connecting sandboxed execution, rollout collection, verifier feedback, and model updates.
- **Modular agent training:** Supports interchangeable agents, environments, and learning algorithms within a shared training and evaluation workflow.

2026 **Matryoshka Agent: learned orchestration for long-horizon MLE** [Paper]

- **Hierarchical agent pipeline:** Helped connect high-level orchestration, sub-agent execution, and iterative ML experimentation for long-horizon tasks.
- **Small-model orchestration:** The framework enabled a **Qwen3-4B** orchestrator to approach o4-mini performance and improved Qwen3-30B-Coder by **up to 36.7% relative** on the evaluated MLE tasks.

2025 **Matryoshka Pilot: learned guidance for black-box LLMs** [Paper] [Code]

- **On-policy harness learning:** Developed a lightweight controller that learns multi-turn guidance from on-policy preference feedback to steer black-box LLMs.
- **Black-box adaptation:** Improved reasoning, planning, and personalization without updating the underlying model's weights.

2025 **MLE-Dojo: interactive environments for ML agents** [Paper] [Code]

- **Executable environments:** Helped build the environment framework and contributed task examples for iterative ML experimentation, debugging, and solution refinement.
- **Training and evaluation at scale:** The framework covers **200+ Kaggle competitions** with executable feedback; evaluations of **eight frontier LLMs** identified limitations in long-horizon problem solving.

## PUBLICATIONS & PREPRINTS

* Equal contribution. † Equal second authorship.

### *Harness Evolution*

[1] **Matryoshka Agent: Unfolding Sub-Agents for Long-Horizon Machine Learning Engineering**
Rushi Qiang, **Changhao Li**†, Haotian Sun†, Yuchen Zhuang, Chao Zhang, Bo Dai.
**Preprint, 2026** [Paper]

[2] **Matryoshka Pilot: Learning to Drive Black-Box LLMs with LLMs**
**Changhao Li***, Yuchen Zhuang*, Rushi Qiang, Haotian Sun, Hanjun Dai, Chao Zhang, Bo Dai.
**NeurIPS 2025** [Paper] [Code]

### *Efficient On-Policy Learning*

[3] **Revisiting DAgger in the Era of LLM-Agents**
**Changhao Li**, Rushi Qiang†, Jiawei Huang†, Chenxiao Gao†, Chao Zhang, Niao He, Bo Dai.
**Preprint, 2026** [Paper] [Code]

[4] **Exploration-Driven Optimization for Test-Time Large Language Model Reasoning**
**Changhao Li**, Yuchen Zhuang, Chenxiao Gao, Haotian Sun, Rushi Qiang, Chao Zhang, Bo Dai.
**TMLR 2026** [Paper]

### *Environments & Rubrics*

[5] **DREAM: Deep Research Evaluation with Agentic Metrics**
Elad Ben Avraham*, **Changhao Li***, Ron Dorfman*, Roy Ganz, Oren Nuriel, Amir Dudai, Aviad Aberdam, Noah Flynn, Elman Mansimov, Adi Kalyanpur, Ron Litman.
**ACL 2026 – Oral** [Paper]

[6] **MLE-Dojo: Interactive Environments for Empowering LLM Agents in Machine Learning Engineering**
Rushi Qiang*, Yuchen Zhuang*, Yinghao Li, Dingu Sagar V K, Rongzhi Zhang, **Changhao Li**, Ian Shu-Hei Wong, Sherry Yang, Percy Liang, Chao Zhang, Bo Dai.
**NeurIPS 2025** [Paper] [Code]

### *Model Efficiency*

[7] **Task-Specific Zero-shot Quantization-Aware Training for Object Detection**
**Changhao Li***, Xinrui Chen*, Ji Wang*, Kang Zhao, Jianfei Chen.
**ICCV 2025** [Paper] [Code]

[8] **Training Transformers with 4-bit Integers**
Haocheng Xi, **Changhao Li**, Jianfei Chen, Jun Zhu.
**NeurIPS 2023** [Paper] [Code]

## SELECTED HONORS

**Outstanding Graduate Award**, Tsinghua University 2024
**Comprehensive Excellence Scholarship**, Tsinghua University 2023
**Academic & Social Service Excellence Scholarship**, Tsinghua University 2022

## ACADEMIC SERVICE

**Reviewer:** NeurIPS, ICLR, ICML, ACL Rolling Review, TMLR.
**Teaching Assistant:** CX4240 – Computing for Data Analysis, Georgia Tech Spring 2026

## TECHNICAL SKILLS

**Languages & frameworks:** Python, C++, PyTorch, CUDA.
**Systems experience:** Implemented CUDA kernels for 4-bit transformer training (NeurIPS 2023).